

COLUMN 1

Time & Freshness

Is evidence still physically valid?

Governing Physics:

- Age: $\Delta t = t_{\text{act}} - t_{\text{sens}}$
- Deadline: $\Delta t \leq \tau_{\text{world}}$
- Drift: $\sigma(t) = \sigma_0 + v\Delta t$

Systems Invariants:

- Freshness Wall Δt_{wall}
- Tail Latency ($P_{99.9}$)
- Latest-Frame Leases
- Multi-Rate Decoupling

Physical Consequence:

Phase lag erosion; unguided open-loop motion.

COLUMN 2

Inertia & Stopping

Can the mass stop within clearance?

Governing Physics:

- Momentum: $\mathbf{p} = m\mathbf{v}$
- Work-Energy: $E_k = \frac{1}{2}mv_0^2$
- Decel Limit: $a_{\text{max}} \leq \mu g$

Systems Invariants:

- Stopping: $v_0\Delta t + \frac{v_0^2}{2a_{\text{max}}}$
- Dynamic Speed Governor
- Barrier Function $h(\mathbf{x}) \geq 0$
- Latency Parameterization

Physical Consequence:

Kinetic overrun; projectile collision with barriers.

COLUMN 3

Actuation Limits

How fast may commands change?

Governing Physics:

- Inductance: $\tau_e = L/R$
- Back-EMF Headroom
- Gear Shear: $\tau \leq \tau_{\text{yield}}$

Systems Invariants:

- Torque Slew Clamp $\|\dot{\tau}\|$
- C^2 Jerk Continuity
- Quintic Spline Interpolation
- Action Chunk Smoothing

Physical Consequence:

Gearbox tooth shear; structural oscillation; trip.

COLUMN 4

Energy & Thermals

What does it cost to sustain action?

Governing Physics:

- Joule Heat: $P = I^2 R$
- Thermal RC: $\tau_{\text{th}} = R_{\text{th}} C_{\text{th}}$
- Burst Scaling: $(I_{\text{cont}}/I_{\text{pk}})^2$

Systems Invariants:

- Firmware $I^2 t$ Clamping
- Winding Temp Limits
- Silicon DVFS Derating
- Cold Bench Budgeting

Physical Consequence:

Stator winding pyrolytic breakdown; DVFS stall.

COLUMN 5

Compute Determinism

Will the silicon finish on time?

Governing Physics:

- Flight Delay: 15 cm/ns
- Memory Bandwidth Wall
- UMA Shared Crossbar

Systems Invariants:

- Zero-Heap: `malloc = 0`
- 1 kHz Static SRAM Loop
- Lock-Free SPSC Queues
- AXI QoS Bus Priority

Physical Consequence:

Tail latency blowout; memory controller stalls.

The Co-Design Invariant: Every physical action proposed by high-level software must simultaneously clear all five physical columns before gate drivers energize motor coils.